Cloud Architecture

Arquitecturas en la Nube

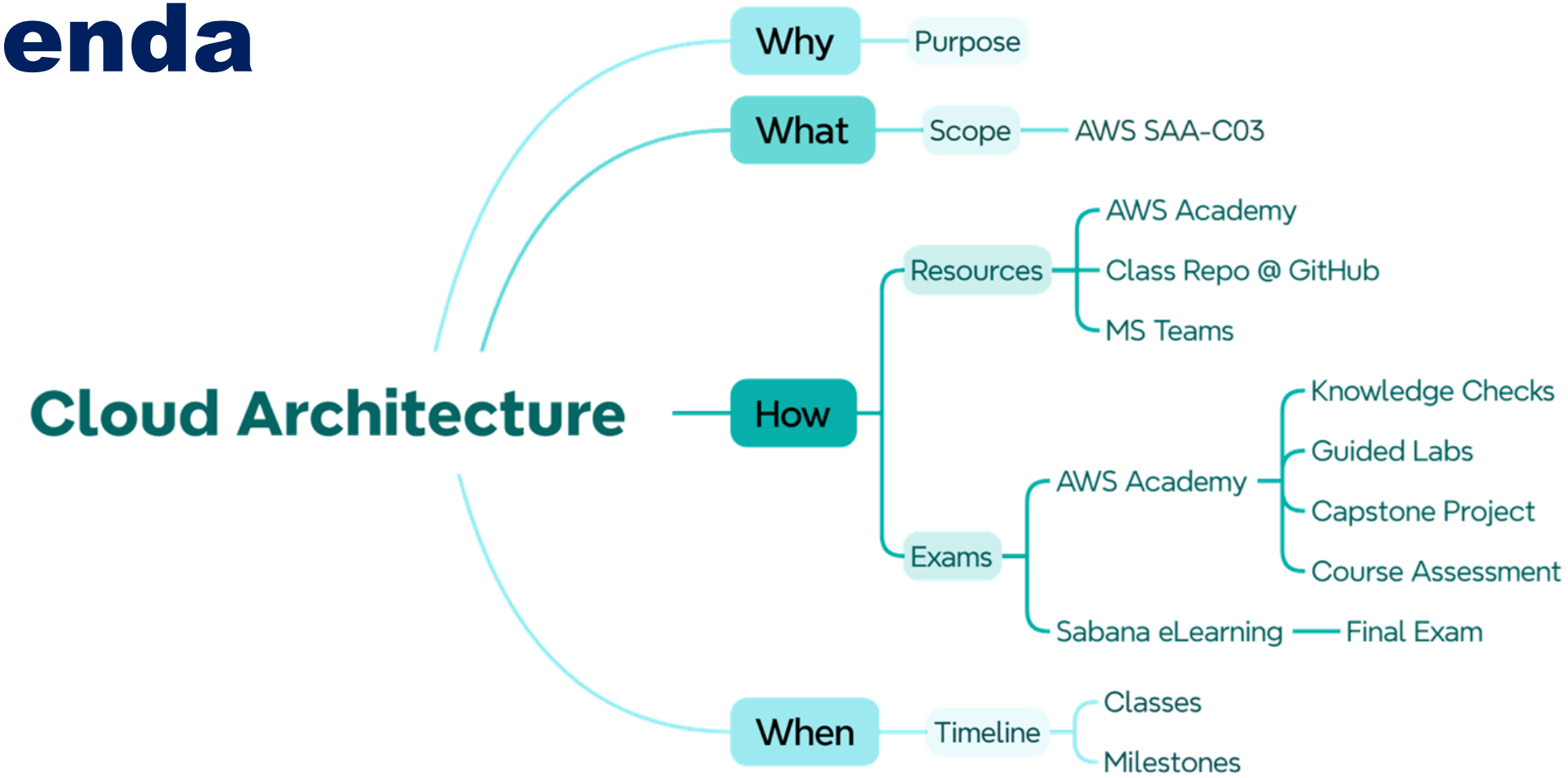


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Agenda



Why & What

Purpose

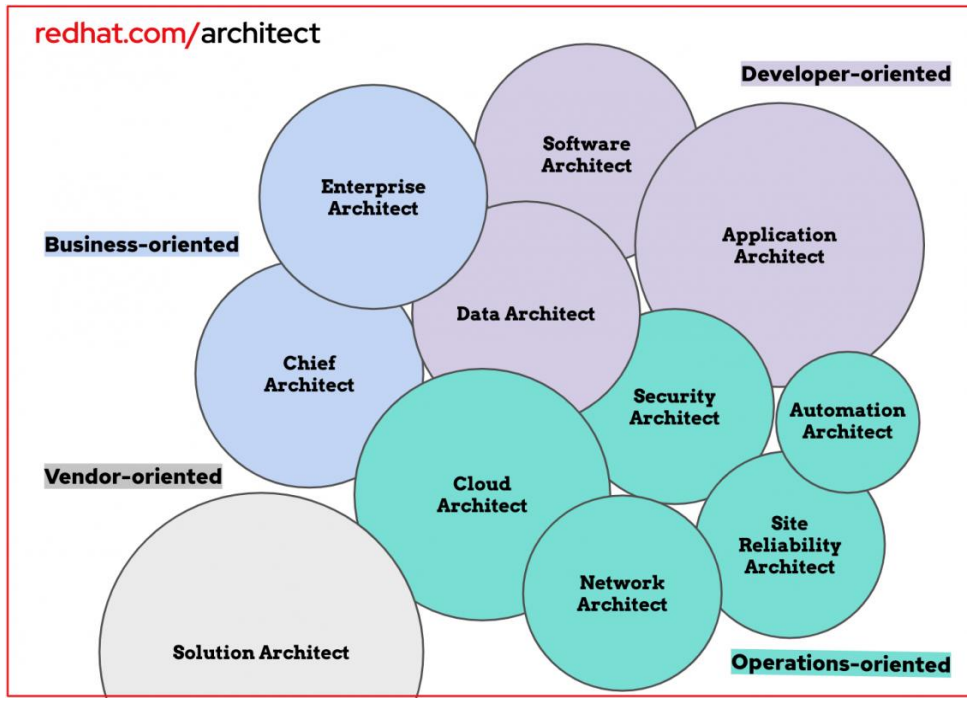
- Use and compare cloud services within one CSP.
- Define your own method to understand the usage of the cloud.

Scope

- Follow AWS Academy Path for AWS SAA-C03 certification
- Understand *Cloud Architect role.*

Architecture Role





Guide Implementation >>
Measure tradeoffs to take decisions

Deliver blueprint/diagram
Describe added business value.

Options: RedHat, TOGAF, SEI

You know the offer (services) on cloud, to apply on architectures defined by a process to build software. You can follow templates (architecture references) and customized/tunning based on your requirements/attributes.

Why not another cloud ?



Market share

Using Gartner MQ or Forrester Wave to review the candidates.

However, you can read an honest review

Employability

Review the number of opportunities on Colombia and compare with others clouds.

Resources available

- Materials
- Labs (during the course)
- Grading Platform
- Paid Training Platform (1 year, Skill Builder)
- Vouchers (4x)

P: Easy to learn

No matter if this is your second or third cloud, you can find common elements; however, the ADN of the CSP will differ when they produce their products.

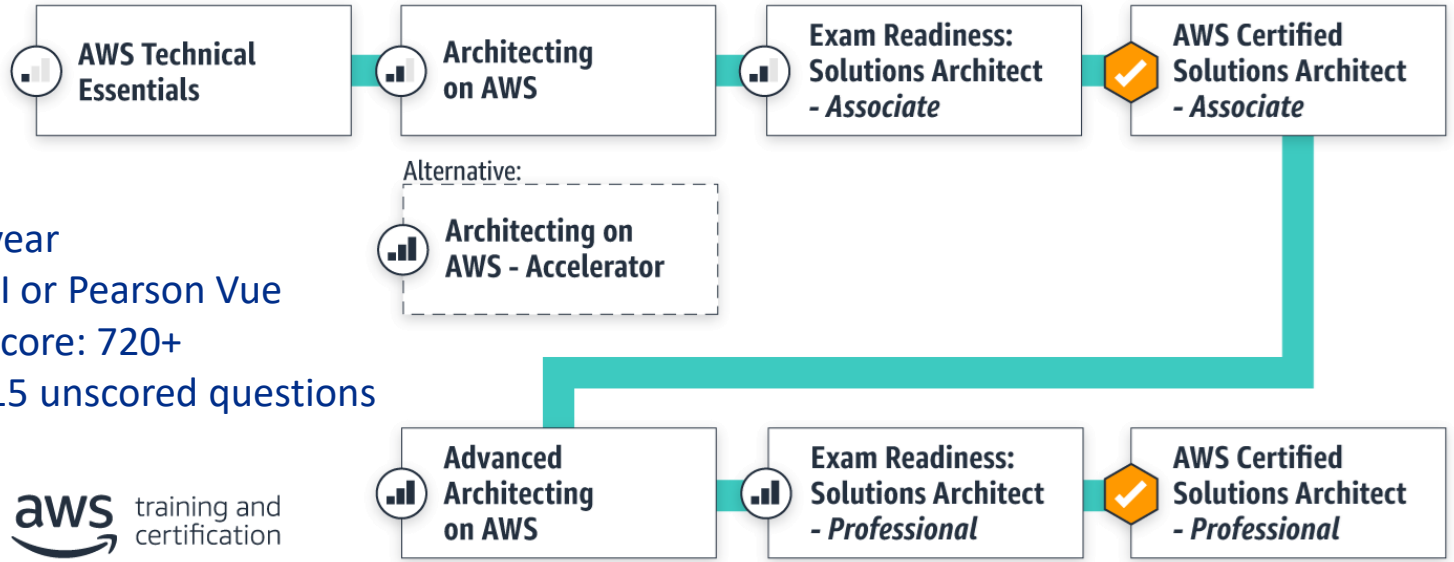
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AWS Architecting Plan



Architect Learning Path

= certification = intermediate
= foundational = advanced



aws training and certification

Add on free digital training at aws.training

SAA-C03
Validity: 3 year
Centers: PSI or Pearson Vue
Minimum score: 720+
50 scored-15 unscored questions

Exam Details

Format
Multiple choice, multiple answer

Type
Associate

Delivery Method
Testing center or online proctored exam

Time
130 minutes to complete the exam

Cost
150 USD (Practice exam: 20 USD)

Language
Available in English, Japanese, Korean, and Simplified Chinese

Domain	% of Exam
Domain 1: Design Secure Architectures	30%
Domain 2: Design Resilient Architectures	26%
Domain 3: Design High-Performing Architectures	24%
Domain 4: Design Cost-Optimized Architectures	20%
TOTAL	**100%**

Approx. 2 mins per question

How: Resources

01 **AWS Academy**

02 **Class Repo@GitHub**

03 **Unisabana e-Learning**



Note: Take advantage of AWS Academy promos and Professor tips for certifications.

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How: Exams

Item	%	Deadlines	On
Capstone Project	10	Last Class	AWS Academy
Knowledge Checks and Course Assessment	25	2 milestones	AWS Academy
Final Exam	30	Last Class	Uni e-Learning
Guided Labs (No Challenge Nor Optional)	35	Every milestone	AWS Academy

Estrategias de evaluación

A continuación, le presentamos el peso evaluativo de las actividades de aprendizaje que se requieren cumplir para obtener la certificación:

Actividad	Ponderación
Laboratorio: Capstone Project – Plataforma AWS Academy	10%
Examen: Knowledge Checks y Course Assessment – Plataforma AWS Academy	25%
Examen: Examen Final – Plataforma Unisabana e-Learning	30%
Laboratorio: Guided Labs (No Challenge Ni Optional) - Plataforma AWS Academy	35%
Total	100%

AWS Academy issues: Create support the ticket directly and keep inform the professor. AWS Academy will need all screenshots.

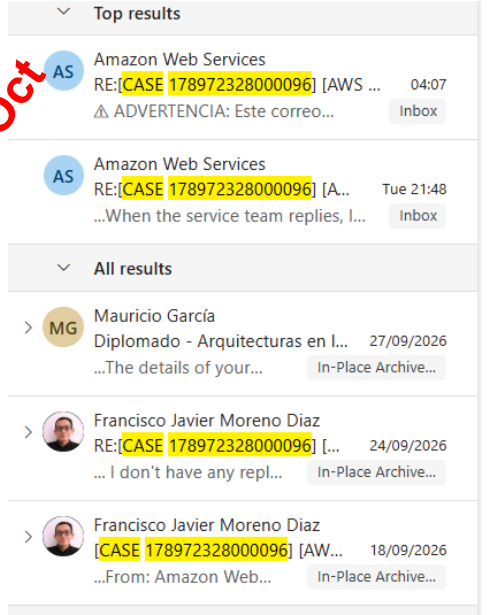
Unisabana e-Learning issues: Review with University Point-of-Contact, the IT support.

Extra Time with regular process only: Review with University Point-of-Contact, the procedure and evidences.

AWS Academy Tips: Capstone Project ask to upload diagram and cost estimation; however, it doesn't apply here. You can review your progress and scores on "Grades" section on the course.

You can do multiple times the labs during the course, review the grade reported on the system if you are motivated by the score.

30th Oct



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When: Timeline

* Review the exact dates for milestone (1,2) on Sabana e-Learning

Class #	Date	Estimated Class according AWS Module	Milestone
1	30-Oct	0,1,2,3,17	
2	3-Nov	4,5	
3	10-Nov	6,7,8	~ Deadline for Mod. 1-8. ~ **Depends how the course run**
4	17-Nov	9,10,11	
5	24-Nov	12,13,14	
6	1-Dic	14,15,16	~ Deadline for Mod. 9-17

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Info from AWS

1. Welcome to AWS Academy Cloud Architecting
2. Introducing Cloud Architecting
3. Securing Access
4. **Adding a Storage Layer with Amazon S3**
5. **Adding a Compute Layer Using Amazon EC2**
6. **Adding a Database Layer**
7. Creating a Networking Environment
8. Connecting Networks
9. **Securing User, Application, and Data Access**
10. **Implementing Monitoring, Elasticity, and High Availability**
11. **Automating Your Architecture**
12. Caching Content
13. Building Decoupled Architectures
14. Building Serverless Architectures and Microservices
15. **Data Engineering Patterns**
16. **Planning for Disaster**
17. **Bridging to Certification**

Component	Number of Occurrences	Estimated Duration in Minutes
Modules (lecture)	17	1,380
Guided labs	14	630
Challenge (café) labs	7	560
Demonstrations	18	200
Activities	8	160
Sample exam questions	14	70
Knowledge checks	15	150
Course assessments	1	60
AWS Academy Cloud Architecting Capstone	1	240
Total estimate in minutes		**3,450**

Module	Lecture	Guided Labs	Challenge Labs	Demonstrations	Activities	Knowledge Checks and Sample Exam Questions
1	1					
2	1					1
3	1	1			1	1
4	1		1	3	1	1
5	1	1	1	2	1	1
6	1	1	1	1		1
7	1	1	1	1	1	1
8	1	1			1	1
9	1	2				1
10	1	1	1	5		1
11	1	1	1	4		1
12	1	1				1
13	1	1				1
14	1	2	1	2	1	1
15	1				2	1
16	1	1				1
17	1					
Totals	**17**	**14**	**7**	**18**	**8**	**15**

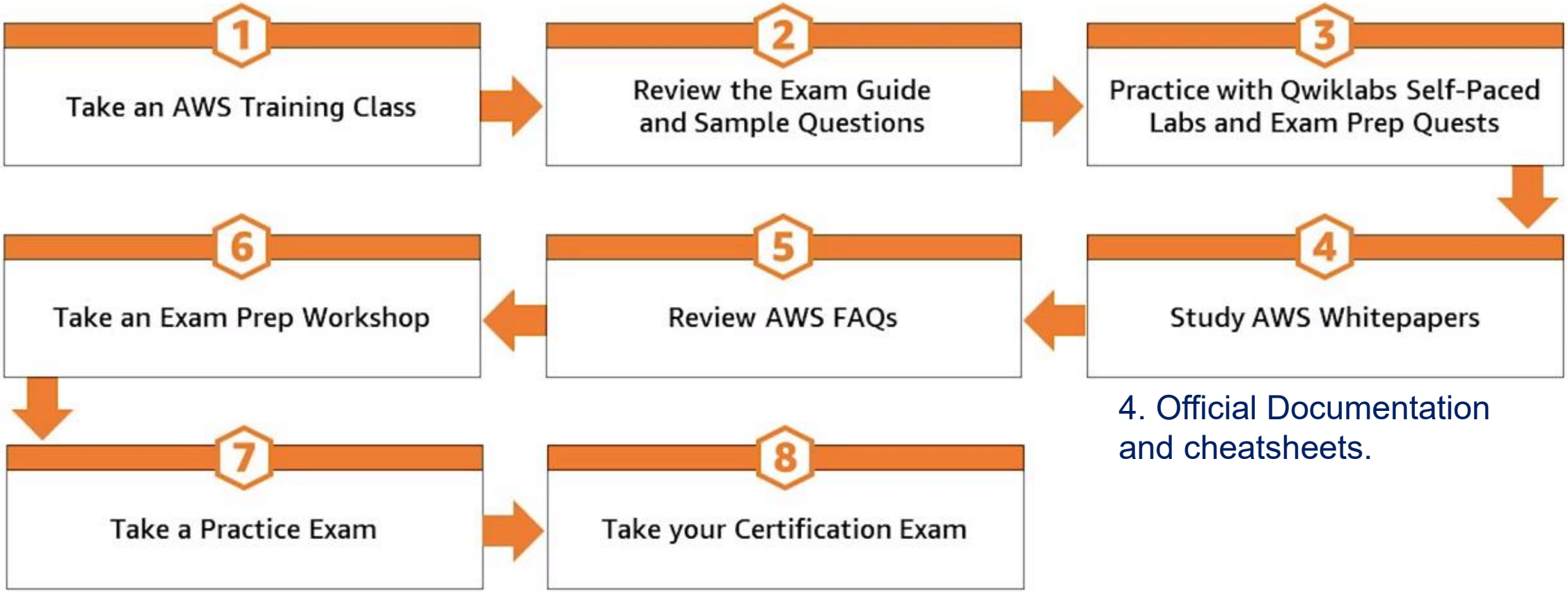
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Recommended Study Plan

2. Official Course: Skillbuilder or 3rd Party courses

1. Objectives/Plan of the exam

3. AWS Builder Labs or AWS Educate Labs.



4. Official Documentation and cheatsheets.

5. Mock Exams on the course or 3rd Party Exams. Loop on 2 or 4.

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Detect your weakness



Amazon VPC

VPN vs Direct Connect



Amazon Simple Storage Service

Bucket Policies vs ACL
S3 Encryption
Storage Types
Lifecycle
S3 Events
Signed URLs



Amazon Kinesis

Kinesis: Streams vs. Firehose vs. Analytics
Kinesis vs. Redshift vs. EMR (you will get scenario-based questions with these)
Athena's differences from Kinesis Analytics



Amazon RDS

Multi-AZ Rep
Read Replica
Snapshots



AWS Lambda

Limits/timeouts
Events and what services are generally used with Lambda triggers

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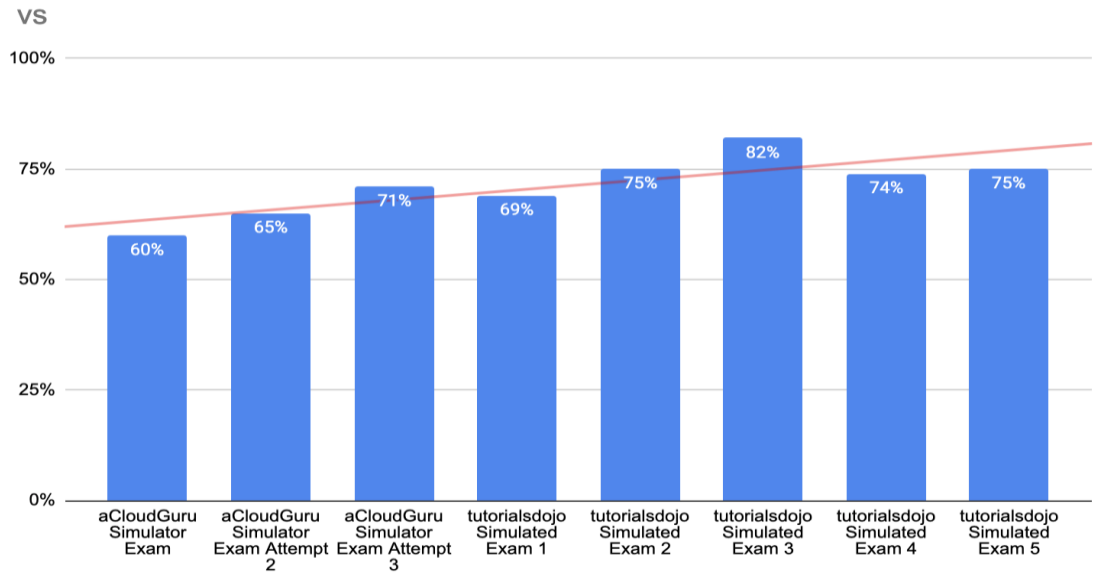
Strategies on the exam

Questions

When presented with a scenario-based question, *read the kicker (Questions) first.*
If they give you information it is relevant to choosing between options, the most of the times.
If they don't give you a piece of information you don't needed it. Don't assume and do not add information that is not there.
Use process of elimination for every question. Look for how the answers self-eliminate
Read questions twice.
Use the “Mark for Review” feature.

Answers

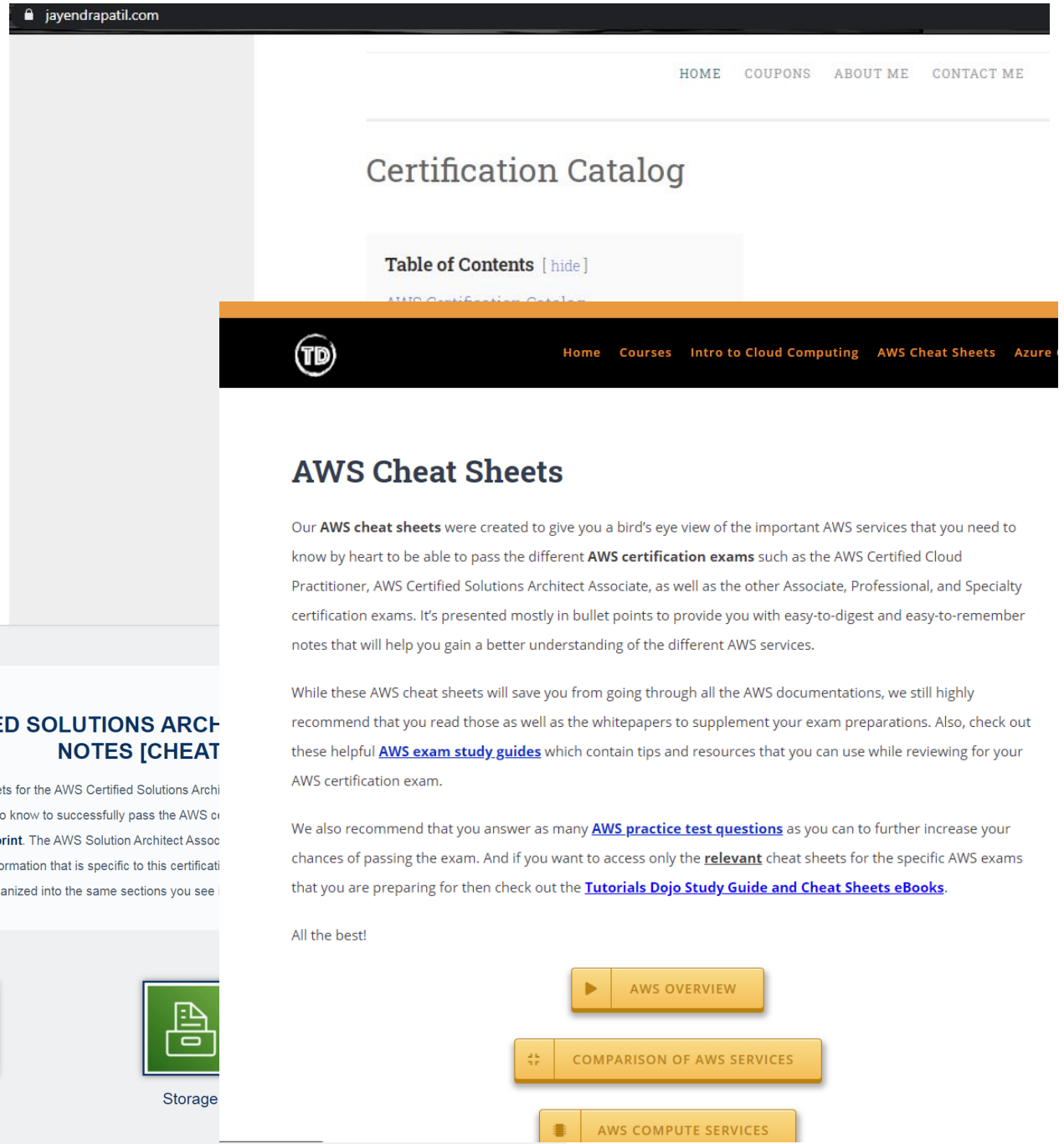
Use questions to answer other questions on the real exam.
You cannot assume additional services or configurations if not given
I suggest that you practice dissecting questions and answers with our practice exams (Dump Exams/Practice Exams).
Do the labs over a few times. but focus on why you are doing each step.



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Cheat sheets

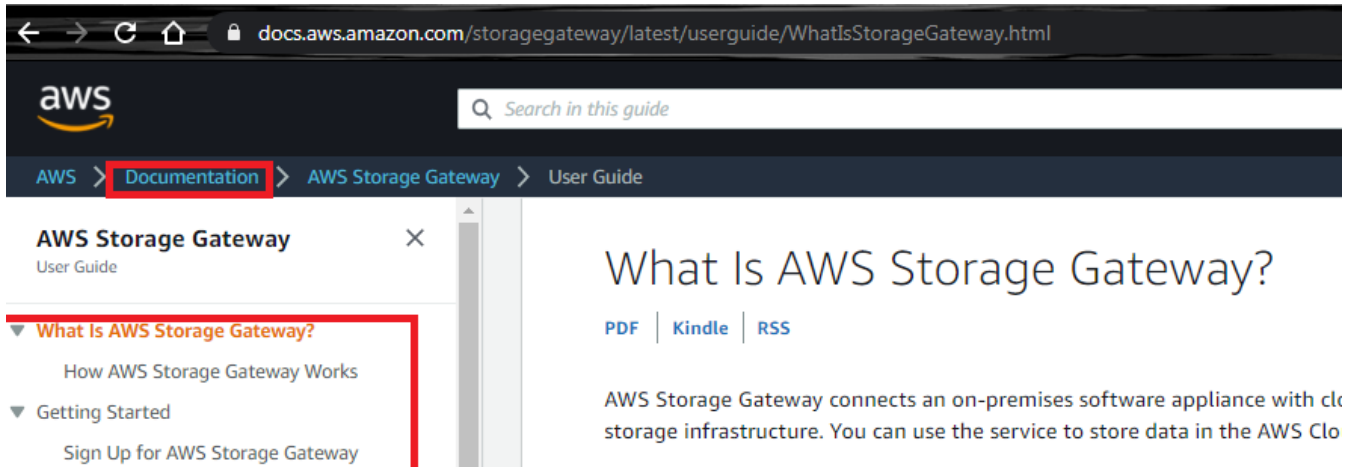
- **All:**
- https://ipspecialist.net
- https://k21academy.com
- **AWS, Azure and GCP:**
- https://tutorialsdojo.com
- https://jayendrapatil.com
- **AWS:**
- https://digitalcloud.training
- https://github.com/open-guides/og-aws

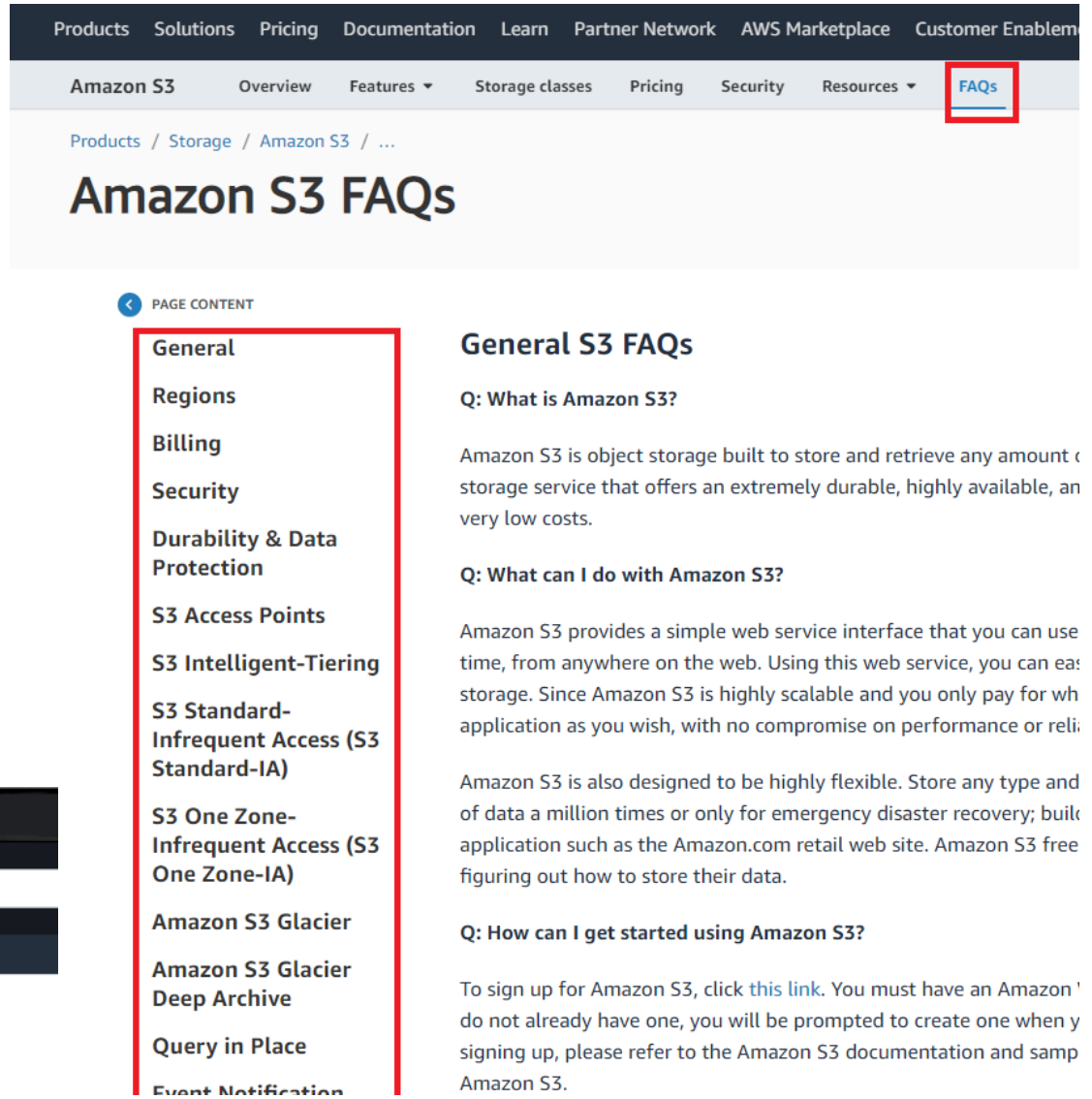


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Detect your weakness

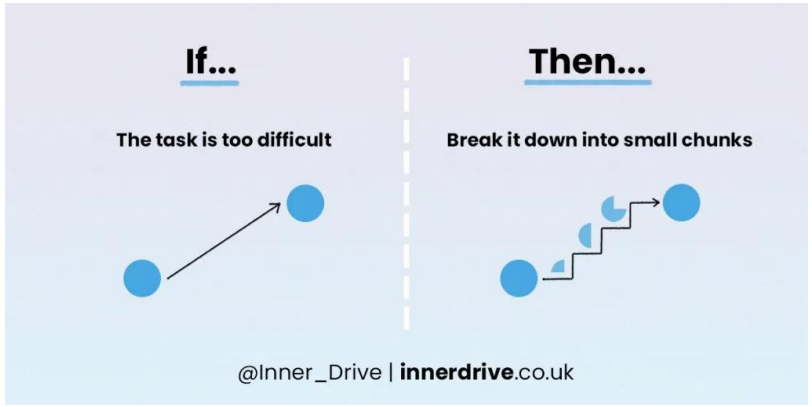
Overview of Amazon Web Services (90 Pags):
https://d1.awsstatic.com/whitepapers/aws-overview.pdf
AWS Storage Services Overview (54 Pags):
https://d0.awsstatic.com/whitepapers/AWS%20Storage%20Services%20Whitepaper-v9.pdf
AWS Security Process (97 Pags):
https://d0.awsstatic.com/whitepapers/Security/AWS%20Security%20Whitepaper.pdf
Fault-Tolerant Components on AWS (23 Pags):
https://d1.awsstatic.com/whitepapers/aws-building-fault-tolerant-applications.pdf
AWS Well-Architected Framework (97 Pags):
https://d1.awsstatic.com/whitepapers/architecture/AWS_Well-Architected_Framework.pdf



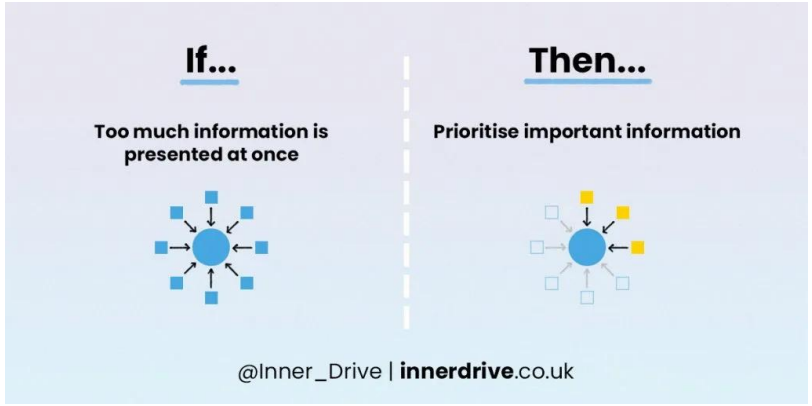


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Cognitive Overloading

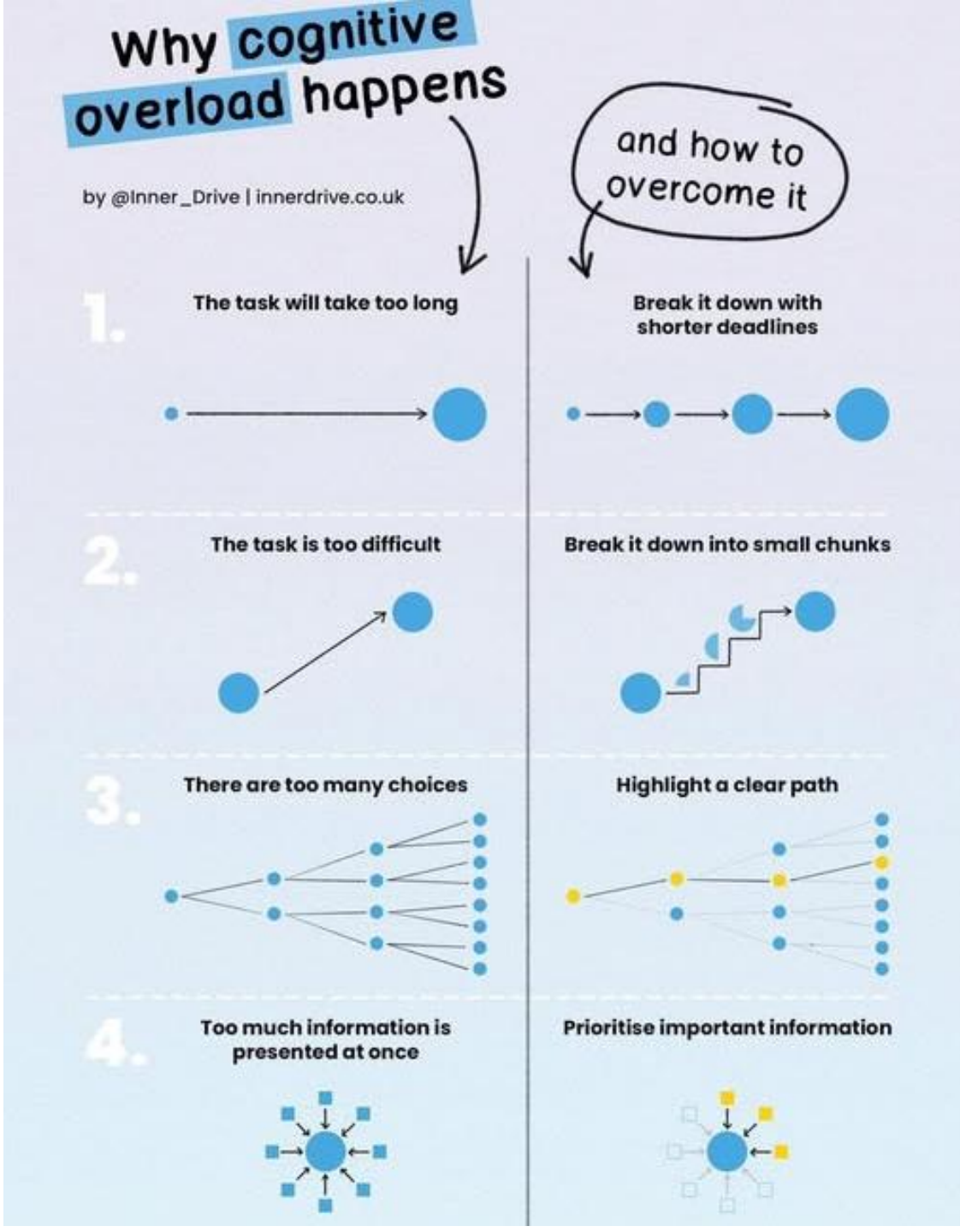


Known what you known, and prepare the unknown lessons



LL: The instructor prioritize slides, and he will focus on cases or experiences.

The instructor skip some slides during the lesson; you have to review the complete presentation on AWS Academy Platform.



- <Topic(s)>

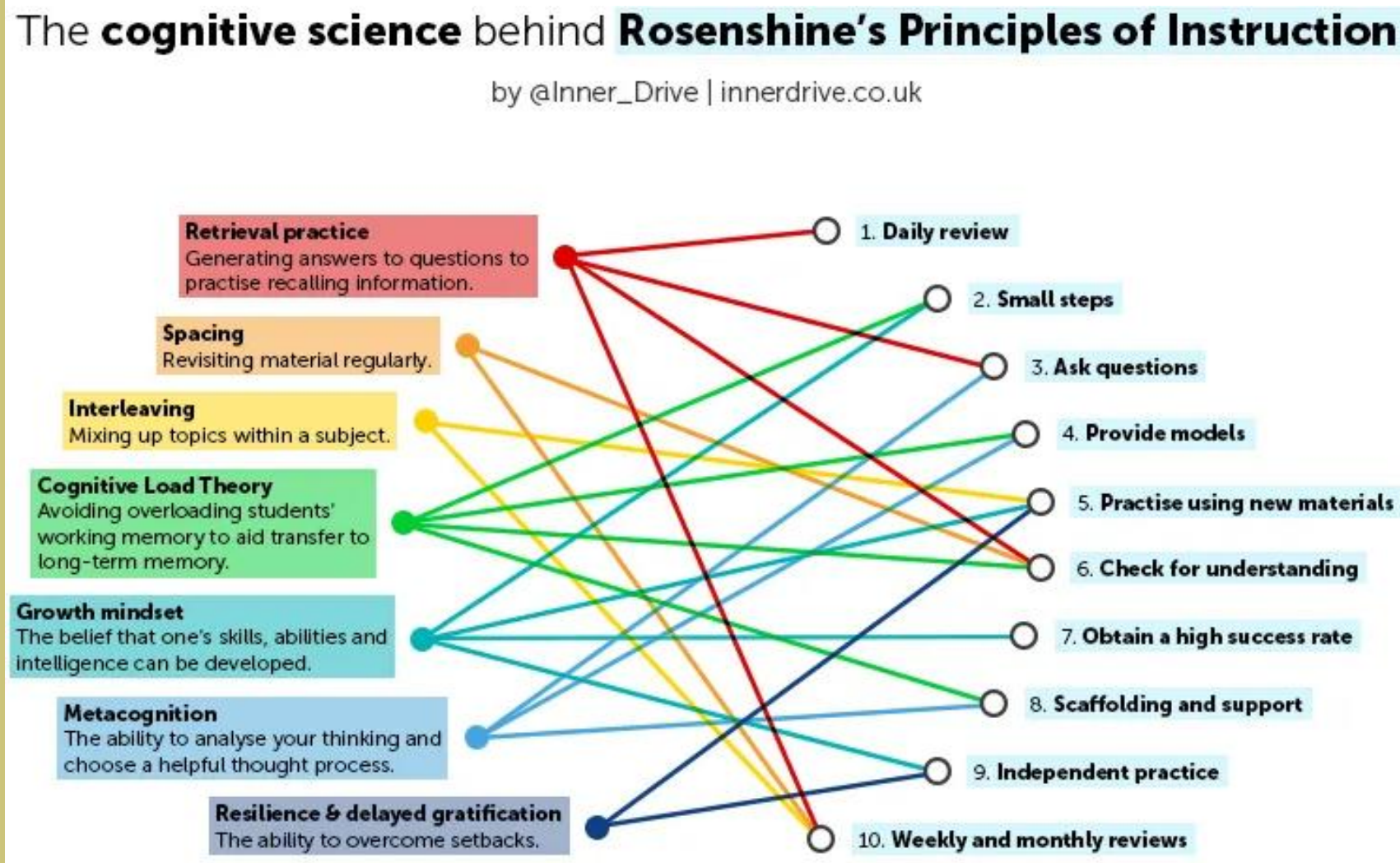


- I don’t have the authority to modify the AWS slides.
- This is the sign (the stamp) for a slide, which:
  - Don’t appear on the AWS Academy course.
  - Give a short story about the usage of the services on the module from instructor experience
  - Suggest a task/research about the service or ideas to student to have in mind.
  - Update about the AWS Services.

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Cognitive Overloading II





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CONTACTO

Francisco Moreno

Profesor Catedra – Depto. Sistemas Inteligentes - Facultad de Ingeniería

franciscomodi@unisabana.edu.co